

V-Wire: A Single-Wire System for Simplified Hardware Prototyping and Enhanced Fault Detection in Education

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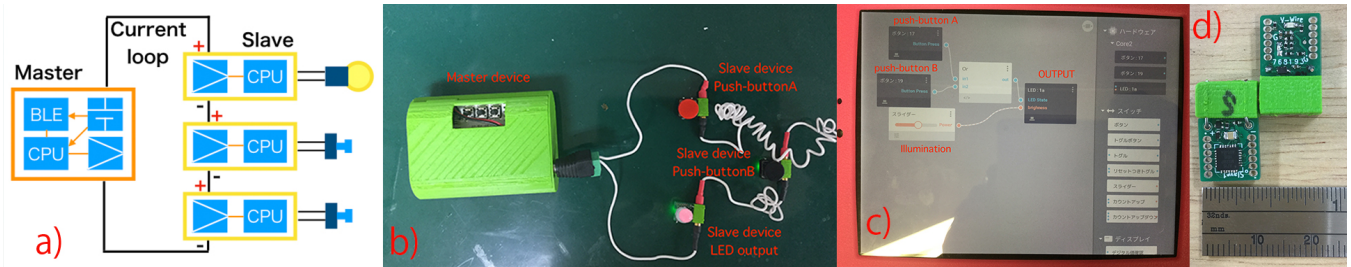


Figure 1: V-Wire system overview: a) Block diagram showing the master device providing two-way communication and power to slave devices connected in series. b) Experimental setup with master device (green box) and various slave devices (push buttons and LED) connected by a single wire. c) Screenshot of the VIVIWARE Cell software interface for visual function connection and control. d) Close-up views of both sides of a slave device circuit board, showcasing its compact design.

Abstract

We introduce V-Wire, an single-wire system designed to simplify hardware prototyping and improve fault detection in educational settings. V-Wire addresses key challenges in hardware development by reducing wiring complexity, eliminating port assignment constraints, and simplifying troubleshooting. The system utilizes a single closed loop wire for two-way communication and power supply, allowing for easy connection of multiple modules and straightforward detection of disconnections. This paper describes the system's architecture, implementation, and its significant advantages in educational applications. This system can support beginner developers in connecting a hardware device easily.

CCS Concepts

• **Human-centered computing** → **Interface design prototyping**.

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1 Introduction and System Overview

The integration of hardware development in education has become increasingly prevalent, yet existing tools like LittleBits [Bdeir and Ullrich 2011] and Arduino [Mellis and Buechley 2012] still present challenges for beginners, particularly in wiring complexity and fault detection. The V-Wire system aims to address these limitations through three primary innovations:

1.1 Single-Wire Design with Serial Connection

V-Wire employs a single closed loop wire for both power supply and two-way communication. Crucially, this design allows for multiple slave modules to be connected in series along this single wire. Each module is simply added to the loop, with its positive terminal connected to the negative terminal of the next module. This serial configuration significantly reduces wiring complexity compared to traditional systems where each module might require separate power and data connections.

1.2 Port Assignment Freedom

The system's ability to easily incorporate multiple slave modules in series enhances port assignment flexibility. This feature allows users to distribute functions across several modules rather than assigning multiple functionalities to a single microcomputer chip. As a result, developers can avoid the complexities of managing multiple functions on limited ports of a single chip, instead utilizing dedicated modules for specific tasks. This approach not only simplifies the design process but also provides greater freedom in prototype modification and expansion, as new functions can be added by simply introducing new modules into the series without the need to reconfigure existing port assignments.

1.3 Simplified Fault Detection

The single-wire, series-connected design allows users to easily detect disconnections by checking the continuity of just one wire. If any module in the series becomes disconnected, it breaks the entire loop, making fault detection straightforward. This feature dramatically simplifies troubleshooting, as users can quickly identify and locate any breaks in the circuit.

1.4 Key Advantages of Serial Architecture

This serial connection of modules along a single wire is a key differentiator of the V-Wire system, enabling simplified construction, flexible expansion, and easy fault detection in educational hardware prototyping scenarios.

2 System Architecture and Implementation

The V-Wire system comprises both hardware and software components designed to simplify hardware prototyping in educational settings.

2.1 Hardware Components

The V-Wire circuit is based on current loop technology, supplying a constant current to power the slave modules while facilitating communication through current changes. The master module detects impedance changes in the loop wire circuit to receive data.

Slave modules (16 x 11 mm) are built around an ATmega328P microcontroller programmable via Arduino IDE, with an estimated additional cost of approximately 2 USD per module. Each slave module is assigned a 48-bit MAC address stored in EEPROM, with the master module discovering slave modules and assigning short local addresses.

2.2 Software Implementation

We have developed the "VIVIWARE Cell" visual programming tool for tablet devices to demonstrate the V-Wire system's capabilities. This application allows users to visually connect and manipulate slave modules, providing an intuitive interface for hardware prototyping and real-time system status monitoring.

This combination of hardware and software components in the V-Wire system creates a comprehensive and user-friendly environment for hardware prototyping in educational settings.

3 Comparison with 1-Wire Technology

While V-Wire shares some similarities with the established famous 1-Wire technology [Analog Devices 2008], there are significant differences:

3.1 Power Supply

1-Wire uses parasitic power, drawing energy from the data line when it's high. V-Wire, in contrast, provides a constant current, allowing for more stable power delivery to modules.

3.2 Topology

1-Wire typically uses a bus topology, while V-Wire uses a closed loop, which simplifies fault detection.

3.3 Power Capacity

V-Wire can provide more power to modules (20mA constant current) compared to 1-Wire's parasitic power, making it more suitable for LED and small actuator applications in educational settings.

3.4 Summary of Key Differences

Given these fundamental differences in power supply, topology, and power capacity, it would be challenging to use 1-Wire technology in the same manner as V-Wire, particularly in educational settings where simplicity, reliability, and the ability to power a variety of modules are crucial factors.

4 Discussion and Future Work

The V-Wire system presents significant advantages in educational settings:

Lowered barrier to entry for hardware prototyping
Intuitive and efficient troubleshooting
Enhanced learning experience through simplified design and fault detection

Current limitations include a maximum of eight slave modules due to voltage constraints, although tests indicate up to 30 modules under experimental setup. The system's 20 mA current limit is suitable for small sensor systems but insufficient for high-power devices.

Future work will focus on increasing module capacity, addressing power limitations for high-consumption devices, and exploring additional educational applications. Potential scenarios for V-Wire application include introductory electronics courses, interactive science exhibits, and rapid prototyping workshops in makerspaces.

5 Conclusion

The V-Wire system offers a promising approach to simplifying hardware prototyping and fault detection in educational settings. By addressing the challenges of wiring complexity, port assignment, and troubleshooting, V-Wire significantly lowers the barrier to entry for students and educators engaging in hardware development projects. The system's unique single-wire design not only simplifies construction but also provides an intuitive and efficient method for detecting and resolving connectivity issues, making it a useful tool for educational environments.

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